An Emotional-Companion Dialogue System on Qwen3-8B: Fine-Tuning, a Separate Profile Reader and Decide-First Decoding

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Code and experiment records: https://github.com/IsemIsCold/Qwenpathy

Abstract

We describe our entry for Topic 2 of the 2026 M-Zone AI+ University Innovation Program: an offline emotional-companion dialogue model. For each conversation it writes the next reply, labels the user's emotion (16 classes) and predicts a user profile (33 tags in three fields). The final system fine-tunes Qwen3-8B with QLoRA, serves the merged weights in FP8 with vLLM on two RTX 4090 cards, makes the model state the emotion before it writes the reply, stops generation after the reply, and takes the profile from a small CPU classifier.

Across five submissions, fine-tuning raised emotion accuracy from 0.58 to 0.82 on 100 held-out points. Moving the profile to the classifier with exact-match thresholds raised per-field exact match from 0.534 to 0.652 on 600 validation points. FP8 serving and early stopping cut mean latency from 1.55 s to 0.90 s. Emotion accuracy, however, stayed between 0.787 and 0.793 across four model versions on those 600 points: every inference-side fix and one re-weighted retrain failed to move it. We trace this ceiling to training labels that disagree on "worried, then relieved" messages. We report all 30 numbered experiments, including the failed ones, each judged by a rule fixed before the run.

Keywords: emotional support dialogue · emotion recognition · user profiling · QLoRA fine-tuning · class imbalance · label noise · low-latency inference

摘要

本文介绍我们参加2026动感地带AI+高校创智计划技术赛道赛题二的方案：一个完全离线运行的情感陪伴对话模型。对每段对话，系统生成下一轮回复，判断用户最新一条消息的情绪（16类），并预测用户画像（3个字段，共33个标签）。最终系统用QLoRA微调Qwen3-8B，合并权重后以FP8精度通过vLLM部署在两张RTX 4090上；模型先给出情绪标签再生成回复，回复结束即停止生成，用户画像则由一个在CPU上运行的小型分类器给出。

在五次提交中，微调将100条留出样本上的情绪准确率从0.58提高到0.82。把用户画像交给分类器并按精确匹配调整阈值后，600条验证样本上的逐字段精确匹配率从0.534提高到0.652。FP8部署和提前停止将平均延迟从1.55秒降到0.90秒。然而，在这600条样本上，四个模型版本的情绪准确率始终在0.787到0.793之间：所有推理端的改进以及一次按类别重新加权的再训练都没能使其提升。我们将这一上限归因于训练标签在“先担心、后释然”类消息上的不一致。文中报告了全部30个编号实验，包括失败的实验，每个实验都按运行前确定的规则判定。

关键词： 情感支持对话 · 情绪识别 · 用户画像 · QLoRA微调 · 类别不平衡 · 标签噪声 · 低延迟推理

1 Introduction

An emotional-companion model has to do three things at once: understand how the user feels, remember who the user is, and answer in a way that fits both. The contest task turns each into a measurable output. Given a Chinese multi-turn conversation that ends on a user message, a system must return the next assistant reply, one of 16 emotion labels for that last message, and a user profile.

The constraints shaped every decision. The submission is a Docker image that the organisers run offline on two 24 GB GPUs, so no external API can be called. One entry script must read the test file and write the predictions with no human step. Latency is scored. A team has at most five successful submissions, and the last one counts. Our compute budget was about 30 GPU card-hours for most of the project.

This paper makes four contributions:

- **A complete, reproducible pipeline** from the official baseline to a fine-tuned, quantised system, with the exact training recipe, serving flags and parsing code of each submission (Sections 4 and 5).
- **A division of labour between two readers.** A CPU classifier decides the slow-moving profile, and the language model decides the emotion first and then writes the reply. This raised the scored profile metric by 12 points of exact match and removed 41% of the generated characters (Sections 4 and 6).
- **A negative result on emotion accuracy.** Prompting, logit corrections, quantisation, a second classifier and a class-re-weighted retrain all leave accuracy at about 0.79 on 600 validation points. We show that the remaining errors sit where the training labels themselves are split (Sections 6 and 7).
- **A record of every idea considered,** tested or not, with the reason for each decision, and the engineering failures that cost us the most time (Sections 7 and 8).

The work was done by a team of first-year and master's students with no prior model-training experience. Our aim was a working entry and an honest account of what we learned.

2 Task, data and scoring

2.1 Task

Each test line holds an id, a conversation id and a history of turns that ends on the latest user message. The system writes one JSON object per line with four fields:

- `response_text`: the next assistant reply, in Chinese.
- `emotion_label`: the emotion of the latest user message, one of 16 labels (joy, gratitude, relaxed, care, pride, neutral, surprise, mixed, sadness, loneliness, anxiety, anger, fear, disgust, shame, helplessness).
- `user_profile`: three tag lists, `personality_traits` (10 tags), `interests` (12 tags) and `style` (11 tags).
- `memory_refs`: memory ids. The test set has no public memory bank, so the official baseline outputs an empty list and so do we.

The organisers run `bash /root/participant/start.sh TEST_FILE RESULT_DIR` inside the public image (vLLM 0.19.0, Ubuntu 22.04, Python 3.12, CUDA 12.8) with no network. An invalid or missing field scores zero for that item.

2.2 Scoring

Table 1. Scoring of the first round, 100 points in total.

Part	Points	Metric
Emotion	25	Accuracy
Reply quality	20	BLEU-4, ROUGE-L and BERTScore against the gold reply [7, 8, 9]
User profile	15	Exact match (confirmed by the organisers on Oct 16)
Latency	10	Mean over the first 100 samples, ranked against the other teams; server start-up excluded
Expert review	30	Integrity, code quality, platform records

2.3 Data

The training set has 8,338 conversations and 18,657 prediction points. The validation set has 900 conversations and 2,354 points. Conversations have a median of 10 to 12 turns and at most 50.

Two properties of the data drive the results in this paper. First, the emotion labels are imbalanced: on validation, anxiety is 31.9% of points, relaxed 17.4% and mixed 7.5%, while surprise, care and joy are each under 1%. Second, most profile fields are empty: interests in 75% of validation points, personality in 62% and style in 52%. An all-empty profile already scores 0.610 per-field exact match on our 600-point slice.

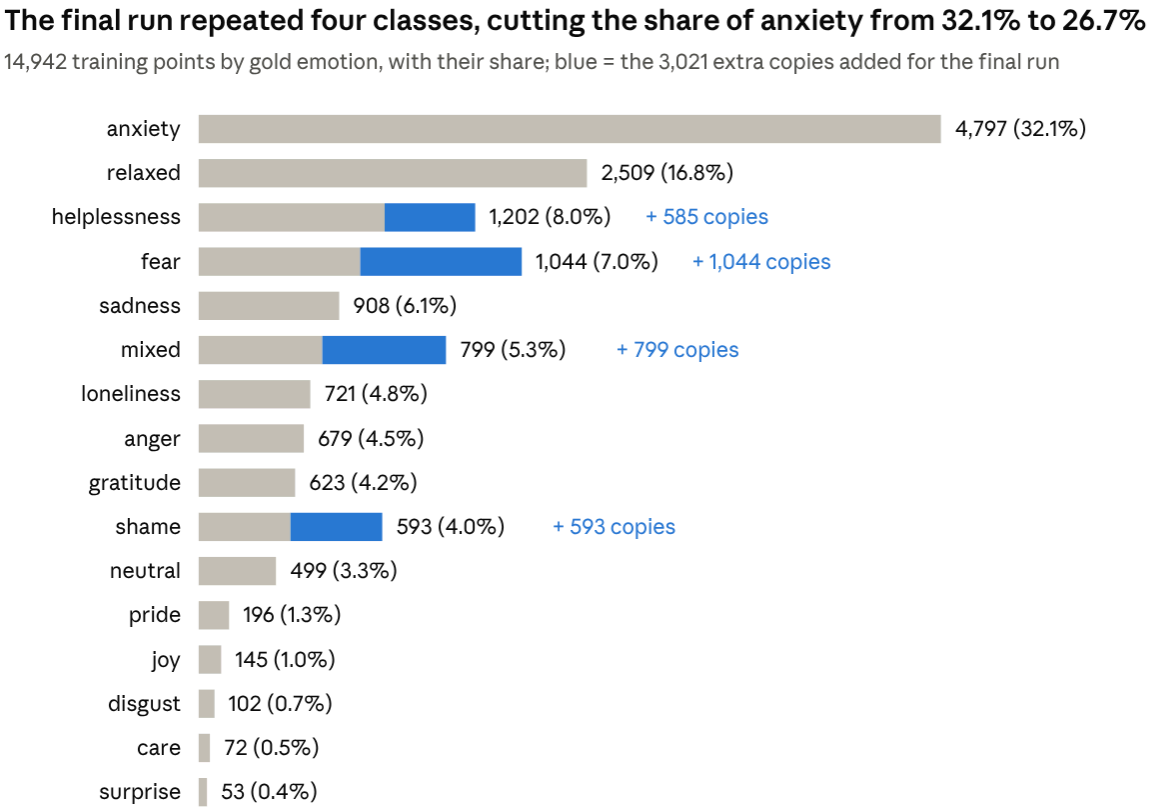


Figure 1 · label counts of the 14,942 training points, from the data build report; extra copies from the data script of the final run

Figure 1 shows the same imbalance in the 14,942 training points we kept (Section 4.2), together with the copies of four classes that the final run added (Section 5.6).

We found one quality problem in the official data. Seven training replies and one validation reply contain the generator model's `<think>` reasoning before the reply. We drop them from training.

2.4 Evaluation protocol

We never trained on validation data. We used fixed slices of it, tracked in the repository:

Table 2. Validation slices used for evaluation.

Slice	Points	Use
n50	50	First 50 points; prompt choices were tuned here

Slice	Points	Use
n50b	50	Random points from other conversations; untouched check for prompt choices
n100 (t9)	100	n50 + n50b; the check run before each submission
tune500	500	Random points outside n100; tuning thresholds
t10	600	n100 + tune500, from 488 conversations; the main comparison set

Three rules governed every comparison. Versions are compared on the same slice, because the same system scores 0.50 on n50 and 0.60 on n50b. Fifty points carry about ±7 points of accuracy noise, so small slices only decide large effects. From the pilot onward, every candidate for submission had a pass rule written down before its run, and a failed rule meant the submitted files were restored automatically.

For differences between two systems we report a 95% bootstrap interval over resampled conversations, and for paired label comparisons the exact McNemar test [58]. Thresholds tuned on validation points are reported both in-sample and with 5-fold cross-validation over conversations. BERTScore was not computed locally, because the platform image has no offline `bert-base-chinese`; reply quality is reported as character-level ROUGE-L and BLEU-4.

3 Related work

This section covers every paper we consulted during the project, grouped by the decision it informed. Section 7 says what we did with each idea.

3.1 Base models, fine-tuning and serving

Our first runs used Qwen2.5 models [2]; the submitted system is built on Qwen3-8B in non-thinking mode [1]. We fine-tune with LoRA [3] on a 4-bit quantised base, as in QLoRA [4], through LLaMA-Factory [5], and serve with vLLM [6]. For latency we considered activation-aware weight quantisation (AWQ) [27] and speculative decoding [28]. An early team plan to combine QLoRA with GainLoRA was dropped when we found no published method of that kind; the idea remains a possible research project of its own.

3.2 Empathetic and emotional-support dialogue

EmpatheticDialogues [10] and ESConv [11] define the two standard framings of the task: replying with empathy to a described situation, and following a support strategy across turns. MoEL [12] decides the emotion first and then generates a reply suited to it, which is

the idea behind our first prompt variant and behind the final decide-first training targets. We listed the Chinese datasets CPED [13], SMILE [14] and CPsyCoun [15] as possible extra training data and did not use them. MemGPT [16] and Generative Agents [17] describe memory for long interactions; our histories fit the context window, so we deferred both. Direct preference optimisation [18] was on our reading list for reply quality and was not reached.

3.3 Class imbalance and emotion benchmarks

A teammate's survey proposed focal loss [19], LDAM [20] and class-balanced loss [21]. Each needs a classification head or a custom trainer, while our model writes the label as text, so the only equivalent open to us was repeating examples by class. The survey also cited MoodBench 1.0 [22]; we checked it and found it is an evaluation benchmark, and that several figures and two citations in the survey could not be verified. We record this because it changed our practice: every cited claim was checked against its source before it influenced a decision.

3.4 Inference-time ideas

We examined self-consistency voting [23], adapter merging with TIES [24] and DARE [25], retrieval of few-shot examples, and automatic prompt optimisation, for which the cited source [26] turned out to address emotion-conditioned generation and not recognition.

3.5 Depth, layers and read-outs

Attention Residuals [29] replaces the residual sum with learned attention over the outputs of earlier layers. Delta Attention Residuals [30] shows that converting a pretrained Qwen3-0.6B this way raises the loss sharply and needs continued pre-training to recover, and it reports that the original gains did not reproduce at larger scale in an independent implementation. DenseFormer [31] reports a similar failure when cross-layer routing is added only at fine-tuning time. We therefore kept only the read-out idea: a learned mix over layers on top of a frozen model, as in the ELMo scalar mix [32].

Probing studies support reading from the middle of the network. A study of Qwen3 models [33] found emotion most separable at about 75% depth, and reported that probes beat prompting by under 2 points on an 8B model. Related work finds sentiment and emotion in early and middle layers of Llama models [34], and that an intermediate layer beats the last layer by 2 to 16% across 32 embedding tasks [35]. Layer-wise attention pooling [36] and a comparison of pooling designs for LLM embedders [37] give mixed evidence that learned pooling beats last-token pooling. On removing layers for speed, deep layers of large models can be pruned with small loss on some benchmarks [38, 39], but newer, post-trained models are less redundant [40].

3.6 Decision models and calibration

Our "System One" plan follows the design of Jev, a commercial model that answers typed questions with calibrated probabilities in one read, placed in front of a generator. No paper describes Jev itself. OpenJev-RLCD [41] is an independent reconstruction: the model writes a short rationale, the distribution over answer tokens is read, and the read-out is trained with a proper scoring rule, optionally followed by reinforcement learning on the rationales. It reports that the first stage carries most of the gain, and that on ChaosNLI [44], where uncertainty comes from annotator disagreement, the objective cannot beat cross-entropy. A separate evaluation [42] found Jev poorly calibrated out of the box and fixed by a threshold fitted on about ten labels, which supports fitting one temperature on held-out data [45]. A practitioner write-up [43] describes the single-token read we already used in our logit-correction experiments. We also checked diffusion language models [46] as a possible faster reader and found no open model that fits the task and our hardware.

3.7 Affect dynamics, personality and social simulation

A master's student shared a reading list on simulating how emotions spread through social networks. From it we took one testable idea, the distinction between fast states and slow traits [47, 48, 49], and one untestable but relevant one, appraisal theory, which separates fear (a certain, near threat) from anxiety (an uncertain, future one) [52]. The list also covered trait models of personality [50, 51], persona-based simulation of users [53], hybrid LLM-plus-cheap-model simulation [54], and network effects [55, 56, 57].

4 Final system

The final entry has two readers and one writer. A CPU classifier reads the user's messages and decides the profile. The fine-tuned language model reads the whole conversation, decides the emotion, and then writes the reply.

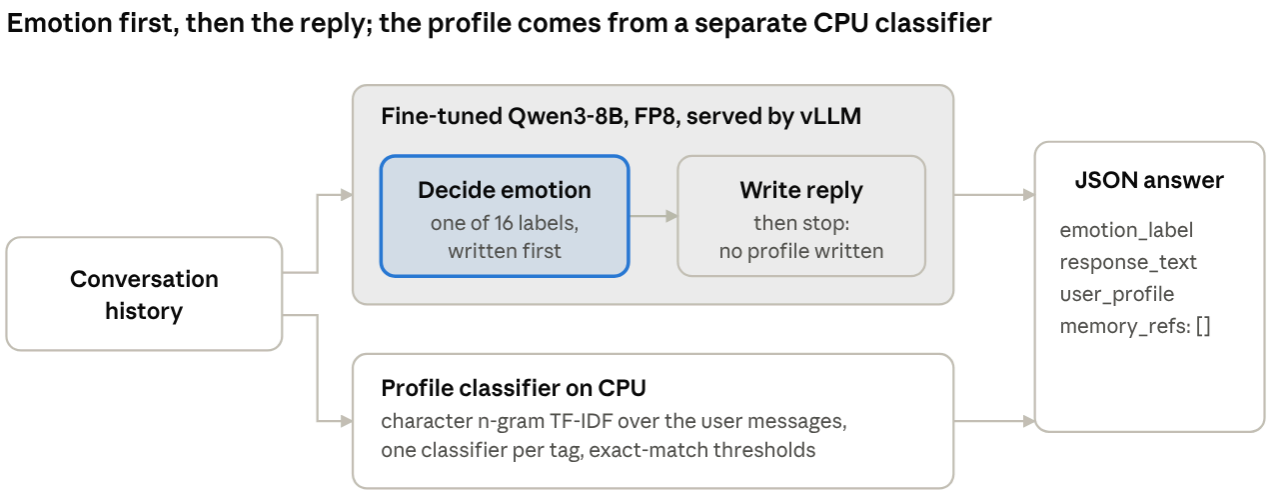


Figure 2 · inference pipeline of the final system

Figure 2 shows the pipeline. The language model never writes the profile, and the classifier never sees the model's output.

4.1 Inference pipeline

1. **Prompt.** The system prompt (V1.8, 355 tokens) lists the 16 emotions and the profile tags in sorted order and asks for one JSON object. The user message is the conversation flattened to `user: …` and `assistant: …` lines. Training and inference import the prompt from the same file, so they cannot drift apart.
2. **Generation.** vLLM serves the merged model with `--quantization fp8`, tensor parallelism over two GPUs, a 32k context, greedy decoding and thinking disabled. The model writes `emotion_label` first and `response_text` second.
3. **Early stop.** The request carries the stop string `"user_profile"`, so generation ends right after the reply. The cut answer is closed with `}` and parsed. If a cut answer lacks a reply or a label, the same sample is generated again in full, inside the timed region. This fallback fired on 0 of 600 validation points.
4. **Parsing.** The parser extracts the first JSON object, maps invented labels to the closest valid one, and retries leniently when a reply contains a raw line break or a stray backslash. That retry recovered 2 of 600 answers that had previously lost both reply and label.
5. **Profile.** A classifier computes character 1-to-3-gram TF-IDF features over the user's messages plus the number of user turns, with one logistic regression per tag, trained on the 18,657 official training points only. A tag is kept when its probability passes a per-field threshold tuned for exact match: 0.45 for personality, 0.30 for interests, 0.35 for style. Inference is plain numpy, about 1 ms per sample, and matches the scikit-learn reference to 1.8e-8.
6. **Output.** One line per input, in input order, with `memory_refs` empty.

4.2 Training recipe

Table 3. Training recipe of the final model.

Setting	Value
Base model	Qwen3-8B, loaded in 4 bits (bitsandbytes)
Adapter	LoRA rank 16, alpha 32, dropout 0.05, all linear layers; 43.6M trainable parameters (0.53%)
Data	14,942 official training points; 2,940 points with gold replies over 350 characters and 7 with leaked reasoning removed; histories cut to 3,500 tokens, oldest turns first
Targets (final version)	JSON in the order emotion, reply, profile, memory; one extra copy of every fear, mixed and shame point and of half the helplessness points, giving 17,963 examples
Optimisation	1 epoch, learning rate 1e-4 with cosine decay, 3% warm-up, global batch 8 on two GPUs, cut-off 6,144 tokens
Template	`qwen3` with thinking disabled, which reproduces the empty think block that vLLM inserts
Tooling	LLaMA-Factory 0.9.5 in a separate virtual environment; adapter merged into full weights for serving
Cost	2,246 steps, about 2 h 10 min on two RTX 4090 cards; dev loss 1.147 at step 250, 0.979 at step 2,000

The first fine-tuned version (submissions #2 to #4) used the same recipe with reply-first targets and no repeated examples: 1,868 steps in 1 h 48 min.

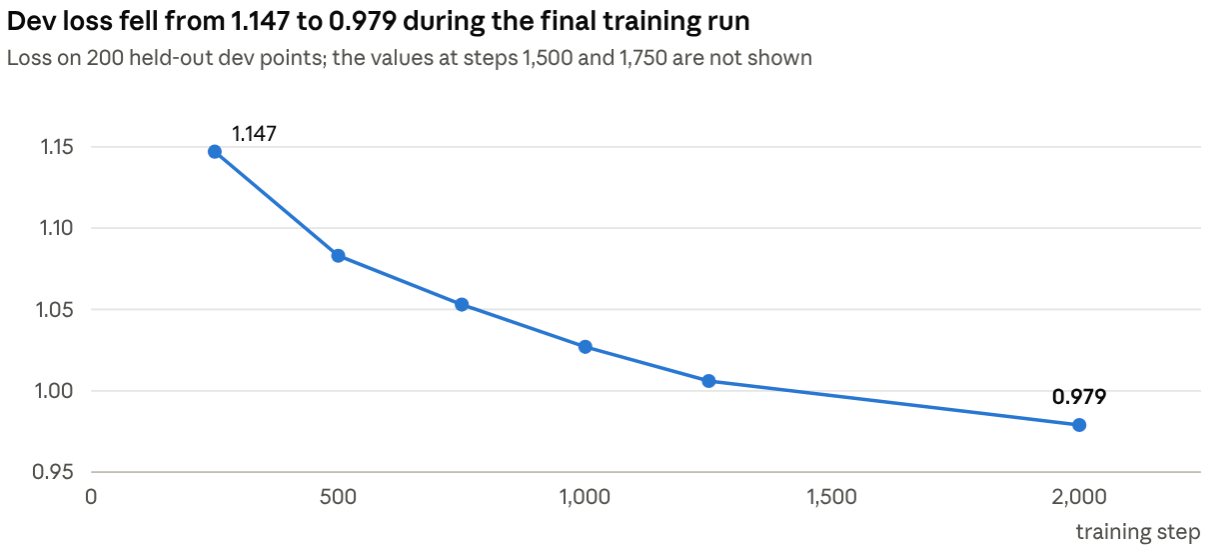


Figure 3 · dev loss of the final training run, from the training log of Nov 10, 2026

Figure 3 shows the dev loss of the final run. It was still falling at the last evaluation, which gives no sign of overfitting within one epoch.

5 Development history

The system went through five submitted versions. Table 4 gives each one's check run: the plain entry script on the same 100 validation points (n100), on the contest hardware. Submission #1 was checked on the first 50 points only.

Table 4. The five submissions and their check runs on 100 validation points (submission #1: the first 50 points).

#	Date	What changed	Emotion	Profile exact	ROUGE-L	Mean latency
1	Oct 4	Base Qwen3-8B, official prompt (V0)	0.48 (50 pts)	not measured	0.173	1.41 s
2	Oct 9	QLoRA fine-tune, V1.8 prompt	0.82	0.477	0.280	1.55 s
3	Oct 16	Same weights in FP8; profile = model tags blended with a CPU classifier	0.84	0.473	0.276	1.14 s
4	Nov 10	Stop after the emotion label; profile from the classifier alone, exact-match thresholds; lenient parser	0.84	0.527	0.276	0.92 s
5	Nov 10	Retrain: emotion decided before the reply, rare classes repeated (V4)	0.82	0.527	0.276	0.90 s

5.1 Baseline and model choice (Oct 3 to 4)

We first ran the official baseline end to end with Qwen2.5-0.5B on a CPU (EXP-001): the pipeline worked and the output was unusable. Qwen2.5-7B through an API gave the right format and generic replies (EXP-002). On 50 validation points, Qwen3-14B scored 0.42 emotion accuracy with thinking on and 0.46 with thinking off, at 2.2 times the latency (EXP-003), and Qwen3-8B tied the 14B model at 0.46 (EXP-004). We chose Qwen3-8B without thinking and submitted it with the official prompt as a pipeline check.

5.2 Prompt phase: V0 to V1.8 (Oct 3 to 5)

Twelve experiments (EXP-005 to EXP-016) changed only the prompt and the parser.

Table 5. Prompt and parser experiments, EXP-005 to EXP-016.

Version	Change	Outcome on 50 points
V0	Official prompt	Emotion 0.50, profile micro-F1 0.275
V1	Emotion field first, base rates, label definitions	No better; the base-rate line pushed fear and relaxed towards anxiety
EXP-007	English prompt	Emotion 0.48; not adopted
EXP-008	Emotion-focused prompt	Emotion 0.52 but anxiety-biased, all profiles empty
EXP-009	Profile-focused prompt	Profile micro-F1 0.329; the idea was kept
V1.3 to V1.5	Reply-length rule, alone and with emotion first	Emotion 0.42 to 0.44; a length instruction hurts emotion and does not move ROUGE-L
EXP-013	Drop three over-predicted tags after generation	Profile micro-F1 0.275 to 0.360
V1.6	"Leave [] if insufficient information" per profile field	Profile exact 0.07 to 0.31; an added definition of *care* caused over-prediction
V1.7	V0 + those profile rules + the tag filter	On the untouched n50b slice: emotion 0.62 against 0.60 for V0, profile exact 0.247 against 0.107
V1.8	V1.7 + mapping of invented labels (about 6% of answers) to valid ones	Frozen for all later versions

The lesson of this phase was negative: no prompt beat the official one on emotion beyond noise, and changing any instruction flipped about 10 of 50 greedy labels. Emotion gains had to come from training.

5.3 Fine-tuning: submission #2 (Oct 6 to 9)

We rebuilt the training data so that each example is exactly the two messages the inference script sends plus the gold answer, then trained in three steps. A 10-step smoke test checked the 4-bit load and the template. A 2,000-example pilot had to pass a gate against the untrained model on both small slices before the full run was allowed; it scored 0.68 and 0.78 against 0.54 and 0.62. The full run on 14,942 examples then reached 0.80 and 0.86. A fresh-boot check with the plain entry script reproduced these within one label, and we submitted.

We also tested a stronger API model, DeepSeek V4.1-Flash, as a possible teacher or data tool (EXP-017). It scored 0.62 and 0.72 on the two slices, below our fine-tuned 8B model, and was not used for training.

5.4 Quantisation and the profile classifier: submission #3 (Oct 12 to 16)

Serving the same weights in FP8 cut latency from 1.55 s to 1.16 s (EXP-020). On its own it lost profile quality on every slice (EXP-023), so it was rejected twice. A CPU classifier for the profile matched the language model alone and, blended with the model's tags, gained 0.035 micro-F1 on 600 points with a confidence interval above zero (EXP-024). FP8 plus the blend beat submission #2 on quality and speed together, and became submission #3 (EXP-025).

5.5 Early stop and exact-match thresholds: submission #4 (Oct 16, submitted Nov 10)

On Oct 16 the organisers confirmed that the profile is scored by exact match. Thresholds tuned for exact match made the classifier alone better than the blend, which meant the language model no longer needed to write the profile at all. Because the model always wrote reply, emotion, profile and memory in that order, stopping after the emotion label removed 41% of the generated characters (EXP-027). The deadline moved to Nov 10 and the team paused; the check run and the submission happened in the final week.

5.6 Decide-first retrain: submission #5 (Nov 9 to 10)

The last version retrained the adapter from the base model with three changes: the emotion comes before the reply in the target, the classes most often mistaken for anxiety are repeated, and a dev loss is logged (EXP-030). It tied submission #4 on every automatic metric (Section 6.2) and failed our own rule for replacing it, which required 12 more

correct emotions out of 600. We submitted it nonetheless as the final entry, as a team decision: at equal scores it is the design we intend to build on, with the emotion decided before the reply and a prediction rate for anxiety that matches the data.

6 Results

6.1 All fine-tuned versions on the same 600 points

Profile and latency improved step by step; emotion and reply quality did not move. Submissions #2 to #4 are re-scored from saved model answers with the parser each one shipped; #5 was run directly.

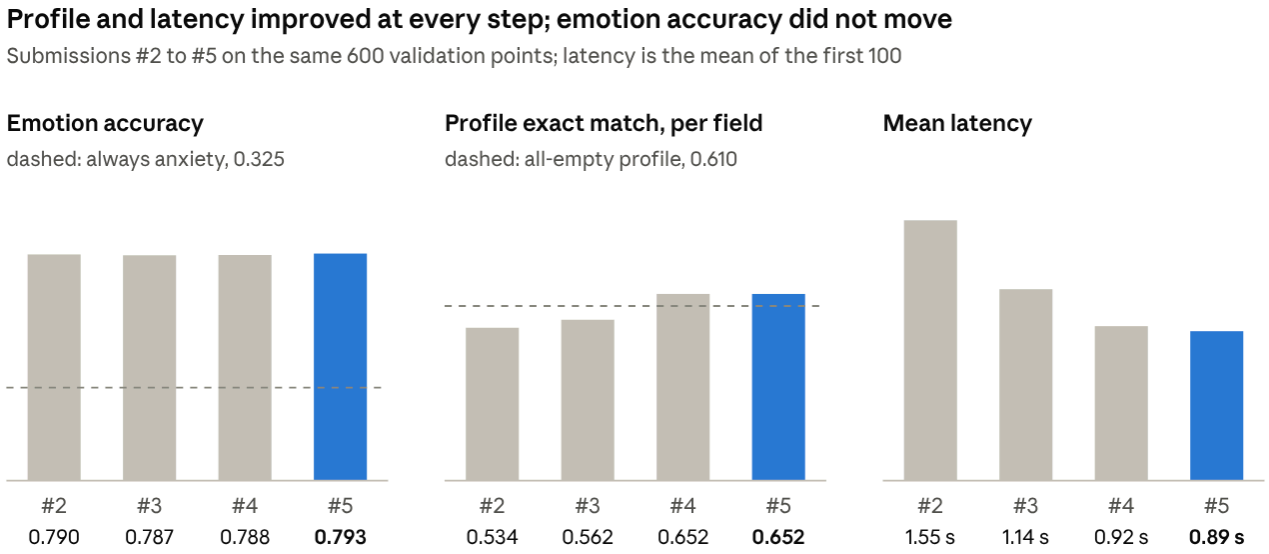


Figure 4 · re-scored from saved model answers on 600 validation points; latency from the check runs

Figure 4 shows the three scored measures side by side. Table 6 adds the counts, the whole-profile score and reply quality.

Table 6. Submissions #2 to #5 re-scored on the same 600 validation points.

	#2	#3	#4	#5 (V4)
Emotion correct, of 600	474 (0.790)	472 (0.787)	473 (0.788)	476 (0.793)
Profile exact match, mean of the three fields	0.534	0.562	0.652	0.652
Whole profile exact (all three fields)	0.220	0.180	0.353	0.353
Reply ROUGE-L	0.294	0.294	0.295	0.296

	#2	#3	#4	#5 (V4)
Mean latency, first 100 points	1.55 s	1.14 s	0.92 s	0.89 s

The profile thresholds of #3 to #5 were fitted on these 600 points. The 5-fold cross-validated estimate for #4 and #5 is 0.647. An all-empty profile scores 0.610 per field and 0.350 for the whole profile, and always answering *anxiety* scores 0.325 on emotion.

6.2 The retrain is a trade, not a gain

Submission #5 and #4 disagree on 75 of the 600 emotion labels: #5 alone is right on 39 and #4 alone on 36. The 95% interval of the accuracy difference, resampling conversations, runs from −0.023 to +0.032. On the 100-point check run the sign reverses (0.82 against 0.84).

Figure 5 explains the tie class by class. Repeating the rare classes did what it was built to do, and cost the same number of points in the two largest classes.

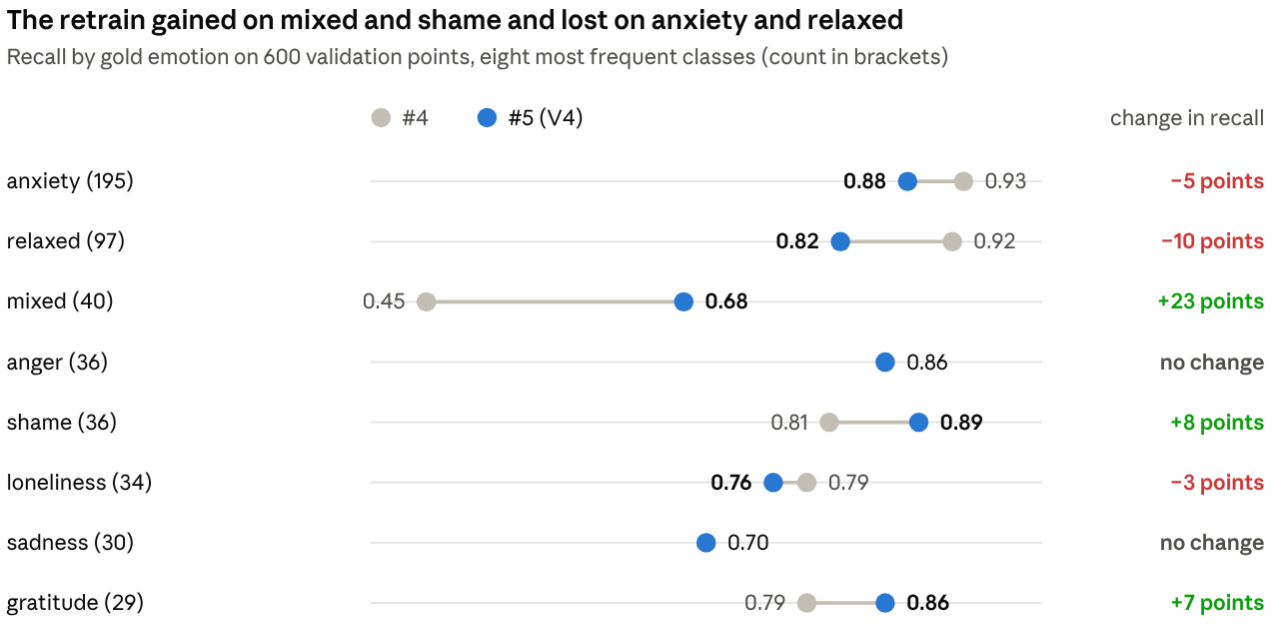


Figure 5 · per-class recall of submissions #4 and #5 on 600 validation points, from the Nov 10 evaluation run

Submission #4 predicts anxiety on 235 points against 195 in the gold labels; #5 predicts it on 203.

6.3 What fine-tuning bought

Fine-tuning is the only change in the project that moved emotion accuracy. Table 7 gives the scores on the two 50-point slices, served with vLLM on the contest hardware.

Table 7. Effect of fine-tuning on the two 50-point slices (n50 / n50b).

	Untrained, V1.8 prompt	Pilot, 2,000 examples	Full run, 14,942 examples
Emotion accuracy, n50 / n50b	0.54 / 0.62	0.68 / 0.78	0.80 / 0.86
Emotion macro-F1	0.238 / 0.336	0.339 / 0.449	0.510 / 0.597
Profile exact match	0.287 / 0.287	0.420 / 0.380	0.453 / 0.500
ROUGE-L	0.174 / 0.197	0.237 / 0.259	0.269 / 0.293
BLEU-4	0.021 / 0.068	0.104 / 0.133	0.130 / 0.151
Median reply length, characters (gold 146 / 100)	85 / 96	128 / 110	150 / 106

6.4 Why emotion accuracy stops near 0.79

Four observations point to the labels, not the model.

- **The gold labels disagree on the confused message type.** For training messages that contain a word for being afraid together with a word of relief, the label is anxiety in 38% of cases and relaxed in 24%. For strong fear words it is fear in 47% and anxiety in 31%.
- **A manual review agrees.** Of the 17 emotion errors of submission #2 on n100, we judged 8 to be model errors, 6 arguable and 3 to be errors in the gold label.
- **The model nearly knows.** For 123 of its 126 errors on the 600 points the gold label is among the model's top 20 alternatives at the label token, with a median gap of 1.12 in log-probability. Yet no re-ranking of those probabilities helped (Section 7).
- **Different models make different errors at the same rate.** FP8 against 16-bit weights: 0.787 against 0.790, with 17 and 15 labels flipped each way. Submission #5 against #4: 39 against 36.

The remaining errors on #3 were dominated by other labels predicted as anxiety: fear 15 times, helplessness 8, mixed 8, shame 6.

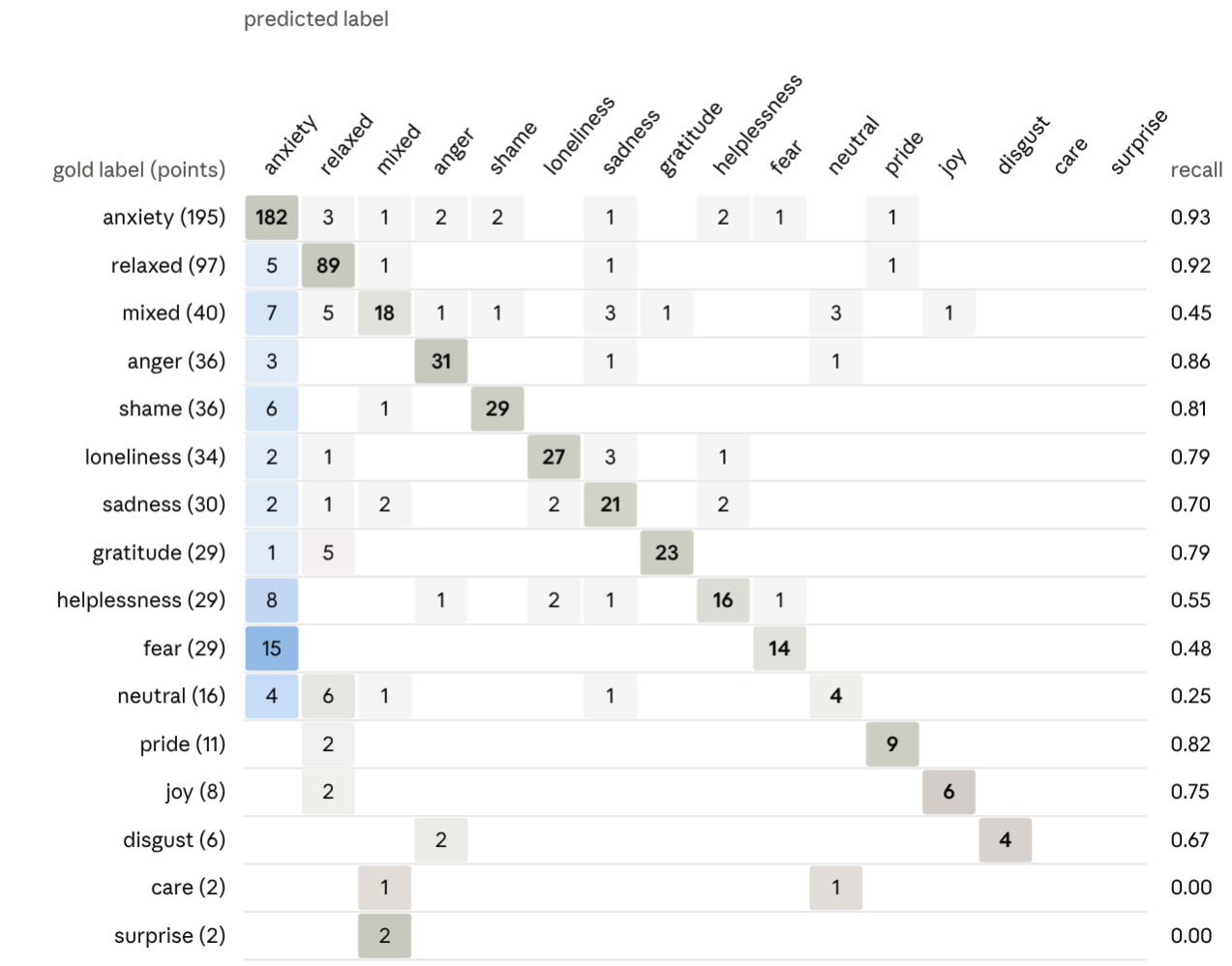


Figure 6 · confusion matrix of submission #4 on 600 validation points, rebuilt from the saved model answers of EXP-023 with the parser of submission #4

Figure 6 gives the full confusion matrix of submission #4, which uses the same weights as #3. Fear is labelled anxiety (15 points) more often than fear (14), and the two classes with no correct answer, care and surprise, have two validation points each. These counts are the reason the final run repeated fear, mixed, shame and helplessness (Section 5.6).

6.5 Why the profile belongs to a separate reader

The profile behaves like a property of the conversation, not of the message. Consecutive prediction points of one conversation carry the same gold profile 72% of the time. A bag-of-character-n-grams classifier over the user's messages matched the fine-tuned language model on micro-F1 (0.468 against 0.461) and beat it on exact match (0.568 against 0.534). With thresholds tuned for the scored metric it reaches 0.652, mostly by leaving a field empty unless it is confident.

6.6 Latency

Each latency step has one cause. Fine-tuning made replies longer and slowed the system from 1.41 s to 1.55 s. FP8 weights brought it to 1.14 s. Stopping before the profile brought it to 0.92 s, 18% below submission #3 measured in the same boot. n-gram speculative decoding made the system 23% slower, because vLLM disables asynchronous scheduling when it is on.

7 Ideas tested and rejected, and plans not run

Most of what we tried did not ship. Table 8 lists ideas we tested and Table 9 lists plans we evaluated on paper or built without running.

7.1 Tested and rejected

Table 8. Ideas tested and rejected.

Idea	Experiment	Result	Decision
Thinking mode	EXP-003, EXP-006	0.42 against 0.46 without thinking, 2.2 times slower; the 8B run was invalid	Off
Larger base model (Qwen3-14B)	EXP-004	Tie at 0.46 on 50 points	8B
Emotion-first prompt with base rates and label definitions [12]	EXP-005, EXP-012	No better; base rates pushed labels towards anxiety	Rejected as a prompt; returned as a training target in V4
Reply-length instruction	EXP-010 to EXP-012	Emotion 0.42 to 0.44; ROUGE-L unchanged; more latency	Rejected
English prompt	EXP-007	Emotion 0.48	Rejected
Stronger API model as teacher or data generator	EXP-017	DeepSeek V4.1-Flash scored below our fine-tuned model; generated data is allowed, but rare classes are 4.3% of validation points	Not used

Idea	Experiment	Result	Decision
Long-reply filter and augmentation study	EXP-018	No training lever worth 1 point; reply length already matches gold	350-character filter kept
FP8 on its own	EXP-020, EXP-023	Profile micro-F1 −2.1 points on 500 points	Rejected alone; accepted with the classifier
n-gram speculative decoding [28]	EXP-021	Same output, 23% slower	Rejected
Anxiety bias from label log-probabilities	EXP-022, EXP-026	+1.4 points in-sample, zero labels changed out of sample; cross-validated +0.007 with an interval spanning zero, +8% latency	Rejected
Per-label offsets on the 16 label log-probabilities	EXP-026	−0.002 and −0.005 cross-validated	Rejected
Removing the tag filter	EXP-022, EXP-026	±0.002 to 0.006 micro-F1	Noise; filter kept
CPU emotion classifier blended with the model	EXP-024	0.643 alone; blend −0.013	Emotion stays with the language model
Self-consistency voting [23]	Simulated from label probabilities	0.778 at temperature 0.5 against 0.790 greedy	Rejected
Emotion inertia from the previous label [47, 48, 49]	EXP-026	−0.003 to −0.005 even with the gold previous label	Rejected
Pooling the profile across a conversation's points	EXP-026	+0.015 only when using the last point, whose history holds the user's future messages	Rejected as a leak

Idea	Experiment	Result	Decision
Opposite-trait constraints [50, 51]	EXP-026	Gold labels are not consistent with trait opposites	Nothing to fix
Profile as a choice among whole tag sets	EXP-028	0.642 against 0.644 per-field exact for independent tags	No gain
Class-re-weighted, decide-first retrain	EXP-030	+3 of 600 on emotion, interval spanning zero	Failed its rule; submitted as #5 by team decision (Section 5.6)

7.2 Considered or built, not run

Table 9. Plans considered and not run.

Plan	Why it was not run
Focal, LDAM and class-balanced losses [19, 20, 21]	Each needs a classification head; our model writes the label as text
Retrieval of few-shot examples	Changes the prompt the model was trained on; needs GPU time to test
Adapter merging with TIES or DARE [24, 25]	We had one real adapter
Automatic prompt optimisation [26]	Train and inference prompts would differ; 50-point slices overfit
AWQ quantisation [27]	FP8 is one serving flag and needs no re-quantisation
Public Chinese dialogue datasets [13, 14, 15]	We trained on the official data only
Continued training of the first adapter	Planned on Oct 16 as the fifth submission; replaced by the full retrain
History truncation at inference	Built and tested, identical to the training rule; switched off because the longest validation histories use about 7,500 of 32,000 tokens
Memory stream and retrieval [16, 17]	Histories fit the context window; deferred to the final

Plan	Why it was not run
Appraisal questions for fear, anxiety and relaxed [52]	Needs teacher labels and a selection study
Attention Residuals retrofit [29, 30, 31]	Published evidence that it breaks a pretrained model without continued pre-training
Layer pruning or early exit [38, 39, 40]	The reply is open generation, the regime most sensitive to pruning
Depth read-out probes [32 to 37]	EXP-029: extraction and probe code built and tested offline (38 checks), with pass rules fixed; the GPU run did not fit before the deadline
Calibrated System One reader, RLCD stage 1 [41, 42, 45]	Planned for the final; stage 2 reinforcement learning is beyond our hardware
Diffusion language models as the reader [46]	The fast ones are API-only; no open one is a stronger reader than Qwen3-8B on our hardware
Newer base model (Qwen3.5-9B)	Needs a new training stack and has no 4-bit path in our tooling; deferred to the final
QLoRA combined with GainLoRA	No published method; a research project in its own right
Rented or university GPUs	Not used; every run was on the contest platform

8 Engineering lessons

These are the failures that cost us the most time or compute, with the rule each one produced.

- **Train with the template you serve with.** LLaMA-Factory's `qwen3_nothink` template does not add the empty think block that vLLM inserts when thinking is disabled. We caught the mismatch by rendering one training example and comparing it with the served prompt, and trained with `qwen3` plus `enable_thinking: false`.
- **Keep one source for the prompt.** The data builder imports the system prompt and the history formatter from the inference script. A test checks that all 18,657 training prompts equal what inference would send.

- **Train in one boot, evaluate in a fresh one.** On our platform the GPUs became unusable within a minute of a training process ending, twice. Chained evaluation scripts died there.
- **Know what is billed.** A boot without GPUs cost one card-hour per hour. Idle sessions, not training, were our largest expense: 23.7 card-hours in six days against 7 planned.
- **Know the image limit.** Uploads failed four times with no error message. The cause was an undocumented rule that the public image plus the used system disk must stay under 50 GB; ours was 99 GB. We moved old models to network storage and removed a 5.3 GB PyTorch install that an early `pip` call had placed in the wrong Python.
- **`nohup` does not protect a distributed launcher.** A dropped terminal sent SIGHUP, and `torchrun` handles that signal itself and exits. Detaching with `setsid` fixed it. Checkpoints every 250 steps let the final run survive three interruptions.
- **Rehearse every GPU session offline.** Each session script was run first against a stub server that replays saved model answers, on both its pass path and its fail path. The one bug that still reached the GPU was a model name the stub did not check.
- **Fix the pass rule before the run.** Every candidate was installed by a script that backs up the submitted files and restores them when the rule fails. Two candidates were stopped this way with the submitted entry untouched.
- **Save raw model outputs.** Saved answers on 600 points let us re-score three submissions without a GPU, test a dozen ideas in seconds, and find a parser bug that had silently dropped 2 of 600 replies.
- **Distrust small slices.** One FP8 run scored 0.76 on 50 points and another 0.82; the 600-point comparison showed no difference. Decisions moved to 500 or more points with confidence intervals.
- **Commit before you close the session.** The first build of the early-stop code was never committed and had to be rebuilt from notes three weeks later.
- **Check every citation.** A survey we were given contained invented class counts and two references that pointed to unrelated papers.

9 Limitations and future work

9.1 Limitations

- **Official scores are not known at the time of writing.** Every number here is from the public validation set.
- **The profile thresholds were tuned on the points they are reported on.** The cross-validated estimate (0.647) is 0.005 below the in-sample figure, but the hidden test set may differ more.

- **Reply quality is measured only by overlap with one gold reply.** We did not compute BERTScore and ran no human or model-based judgement of empathy, which is what the task is about.
- **The final submission did not pass our own replacement rule.** It ties the previous one; we chose it for its design.
- **Memory is handled only by keeping the whole history in context,** and `memory_refs` is always empty.
- **Each configuration was trained once,** with one seed, so we cannot separate the effect of decide-first targets from that of class repetition.
- **The label-noise argument rests on 600 points** and on our own reading of 17 errors.

9.2 Future work

1. **Combine the two fine-tuned models.** They are right on different points at the same rate. Reading the 16-way label probabilities of both and testing a combination needs one short GPU run and no training.
2. **Run the depth read-out study** (EXP-029) on both models: is the emotion easier to read from the middle layers than from the generated label?
3. **Train a calibrated reader.** A classification read-out trained with a proper scoring rule, with one fitted temperature and a measured calibration error, whose probabilities can also shape the reply when the model is unsure.
4. **Ask appraisal questions.** Separate fear, anxiety and relaxed by asking whether the threat is certain, near or resolved, with the questions selected on training data.
5. **Repair the labels.** Use soft labels or relabelling for the message types where the gold labels split.
6. **Compare base models pilot against pilot,** including Qwen3.5-9B, on the same 600 points and with the same rules.
7. **Add memory for long sessions** and an evaluation of empathy with simulated users.

10 Conclusion

A fine-tuned 8B model, a 35 MB classifier and careful serving are enough for a complete emotional-companion entry on two consumer GPUs. Fine-tuning on the official data was the one change that improved emotion recognition, from 0.58 to 0.82 on 100 held-out points. Everything after that improved what the scoring rewards elsewhere: the profile, by giving it to a reader built for a slow-moving signal and tuning it for the scored metric, and latency, by quantising the weights and not generating what another component already knows.

The emotion score did not respond to anything else we tried, and the evidence points to disagreement in the labels on a specific kind of message. The final system therefore changes how the decision is made, with the emotion committed before the reply, without claiming a higher score for it. The next gains will have to come from a reader that knows when it is unsure, or from better labels.

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Appendix A: index of experiments

Every numbered experiment, in order. Each has a folder in the project repository with its goal, set-up, results and decision.

Table A1. Index of the 30 numbered experiments.

ID	Date (2026)	Question	Outcome
EXP-001	Oct 3	Does the official baseline run end to end?	Yes; output of a 0.5B model unusable
EXP-002	Oct 3	What does a 7B model give with the official prompt?	Correct format, generic replies
EXP-003	Oct 4	Thinking or non-thinking mode?	Thinking no better and 2.2 times slower
EXP-004	Oct 4	Qwen3-8B or 14B?	Tie; 8B chosen

ID	Date (2026)	Question	Outcome
EXP-005	Oct 5	Overview of prompt versions V0 to V1.8	No prompt beats V0 on emotion; V1.8 wins on profile
EXP-006	Oct 3 to 5	Thinking mode on 8B	Invalid run; thinking stays off
EXP-007	Oct 3 to 5	English prompt	Not adopted
EXP-008	Oct 3 to 5	Emotion-focused prompt	Anxiety-biased, empty profiles; not adopted
EXP-009	Oct 3 to 5	Profile-focused prompt	Idea kept for V1.8
EXP-010	Oct 5	V1.3: profile and length rules	Emotion worse; rejected
EXP-011	Oct 5	V1.4: length rule only	Emotion worse; rejected
EXP-012	Oct 5	V1.5: emotion field first	No help; rejected
EXP-013	Oct 5	Drop three over-predicted profile tags	Profile micro-F1 0.275 to 0.360; kept
EXP-014	Oct 5	V1.6: leave a field empty if unsure	Profile rules kept, emotion part rejected
EXP-015	Oct 5	V1.7: V0 + profile rules + tag filter	Candidate
EXP-016	Oct 5	V0 against V1.7 on an untouched slice	V1.7 confirmed; V1.8 frozen
EXP-017	Oct 7	Is a stronger API model a useful teacher?	Works; not used for training
EXP-018	Oct 12	Reply length, the 350-character filter, augmentation	No training lever worth 1 point
EXP-019	Oct 8 to 9	QLoRA pilot, full run and submission check	Emotion 0.54 to 0.80 and 0.62 to 0.84; submission #2
EXP-020	Oct 12	FP8 weights for latency	25% faster; quality gate failed on its own

ID	Date (2026)	Question	Outcome
EXP-021	Oct 12	n-gram speculative decoding	23% slower; rejected
EXP-022	Oct 13	Anxiety bias from label log-probabilities; tag filter off	No change out of sample; rejected
EXP-023	Oct 13	FP8 plus anxiety bias as a submission	Profile gate failed; no-go
EXP-024	Oct 13	Can a CPU classifier help on profile or emotion?	Profile blend +0.035 micro-F1; emotion no gain
EXP-025	Oct 13 to 16	Ship FP8 plus the profile blend	Check run passed; submission #3
EXP-026	Oct 16	Ideas from a shared reading list, tested offline	Nothing adopted
EXP-027	Oct 16, Nov 8 to 10	Stop after the emotion label; exact-match thresholds	Check run passed; submission #4
EXP-028	Oct 19	Profile as independent tags or as whole tag sets?	No difference
EXP-029	Nov 5	Depth read-out probes over all layers	Built and tested offline; not run
EXP-030	Nov 9 to 10	Decide-first, class-re-weighted retrain (V4)	Tie with #4; submission #5